PDC Reconstruction Literature Survey and SAMURAI Terminal Air-MS Gap Analysis

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1 TL;DR

This document surveys the published reference frame for the SAMURAI PDC1/PDC2 drift-chamber chain and contrasts it with our current Geant4 reconstruction performance, in order to localise the air multiple-scattering gap that drives our PDC2 spatial resolution one order of magnitude above design.

- **Reference frame.** The PDC1/PDC2 chain has no dedicated peer-reviewed publication. The canonical references are the SAMURAI overview paper by Kobayashi et al., Nucl. Instr. Meth. B **317**, 294–304 (2013) §3.4[1], the RIKEN internal technical note `Detector-PDC.pdf`[2], and the Kahlbow PhD thesis (TU Darmstadt, 2019)[3] which documents an independent reconstruction of the same hardware.
- **Design specifications.** Combining Kobayashi 2013 §3.4[1] with the SAMURAI 2012 construction proposal[4], the PDC1/PDC2 chain targets 1 mm position resolution, 2 mrad angular resolution, and 0.8% relative momentum resolution ($\Delta p/p$). Commissioning RI-beam runs reached $\Delta p/p \approx 1/1500$[1]; the original proton-mode design value is $\Delta p/p \approx 1/700$ as quoted in the RIBF-TAC05 (2005) construction proposal[5].
- **Our measured σ_y(PDC2).** In our Geant4 + reconstruction chain we observe $\sigma_y(\mathrm{PDC2}) =$ 12.83–14.93 mm in the all-air configuration, 6.80–8.21 mm when Region A (dipole cavity + beam pipe) is set to vacuum (`beamline_vacuum`), and approximately 1.0 mm in the all-vacuum configuration[6]. The realistic-air values sit roughly an order of magnitude away from the 1 mm hardware design specification[1].
- **Air MS dominates and the configuration choice matters.** A quadrature decomposition[6] attributes the all-air budget to $\sigma_A \approx 11.19$ mm from Region A (cavity + beam pipe, $\sim$4.3 m) and $\sigma_B \approx 7.43$ mm from Region B (Exit Window $\to$ PDCs, $\sim$2 m). The stage D inputs from the same memo[7] give $\sigma_y(\mathrm{PDC2}) = 7.49$ mm if Region A is vacuum and $\sigma_y(\mathrm{PDC2}) = 13.50$ mm if both Region A and Region B are air. Whether Region A is air or vacuum during the experiment is therefore a configuration question that selects between these two values; this document does not assert which one corresponds to the actual run.

See §6 for proposed action items.

2 The problem: air multiple scattering at the SAMURAI terminal

2.1 Geometry of the proton flight path

The deuteron target sits at world coordinates $(-12.488, 0.0009, -1069.458)$ mm, which corresponds to yoke-local coordinates $(-545.5, 0, -919.9)$ mm under the -30° yoke rotation about Y[8]. Because the SAMURAI dipole cavity (`vchamber_box`) extends over $z \in [-1750, +1750]$ mm

[1]T. Kobayashi et al., Nucl. Instr. Meth. B **317**, 294–304 (2013), DOI: 10.1016/j.nimb.2013.05.089.

[2]RIKEN Nishina Center, *Detector-PDC* technical note, `https://www.nishina.riken.jp/ribf/SAMURAI/image/Detector-PDC.pdf`.

[3]J. Kahlbow, PhD thesis, TU Darmstadt (2019), TUprints 9192, `https://tuprints.ulb.tu-darmstadt.de/9192/`.

[4]SAMURAI Collaboration, *SAMURAI Construction Proposal* (2012), `https://ribf.riken.jp/SAMURAI/120425_SAMURAIConstProp.pdf`.

[5]SAMURAI Collaboration, *RIBF-TAC05 SAMURAI construction proposal* (2005), `https://ribf.riken.jp/RIBF-TAC05/10_SAMURAI.pdf`.

[6]See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §4.

[7]See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §6.

[8]See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §1.1–§1.3 for coordinates, yoke rotation, and Region A/B definitions.

in yoke-local frame, the target at $z = -919.9$ mm is *physically inside the dipole cavity* rather than upstream of it. Two regions therefore separate the target from the PDCs. **Region A** (≈ 4.3 m) covers the in-cavity flight from the target through the cavity exit (2.67 m) and onwards through the downstream beam-pipe (`VacuumDownstream`, ≈ 1.6 m) up to the Exit Window; this region is controlled by the Geant4 macro switch `/samurai/geometry/BeamLineVacuum`. **Region B** (≈ 2.0 m) covers free flight in the world volume from the Exit Window at world $(170.5, 0, 3187.5)$ mm to PDC1 at $(700, 0, 4000)$ mm (0.97 m) and on to PDC2 (1.00 m), gated by `/samurai/geometry/FillAir`. Inside Region A the proton bends $\sim 30°$ in the 1.15 T field, but the arc-length vs. straight-line difference stays below 10%[8].

2.2 Measured σ values

Table 1 reproduces the robust half-width $\sigma = (p_{84} - p_{16})/2$ measured at PDC1 and PDC2 across three vacuum/air configurations and three truth momenta, with $|\vec{p}| = 627$ MeV/c protons.

Condition	(p_x, p_y) [MeV/c]	N	σ_x(P1)	σ_y(P1)	σ_x(P2)	σ_y(P2)	σ_{θ_x}	σ_{θ_y}
all_air	$(0, 0)$	50	3.071	9.736	3.669	12.826	12.279	9.240
all_air	$(100, 0)$	50	3.838	10.492	5.377	12.751	15.768	8.277
all_air	$(0, 20)$	50	3.372	13.039	4.065	14.927	21.709	7.846
beamline_vacuum	$(0, 0)$	50	1.528	4.061	2.086	6.801	11.834	5.910
beamline_vacuum	$(100, 0)$	50	2.067	4.888	3.060	8.211	8.100	5.786
beamline_vacuum	$(0, 20)$	49	1.667	4.789	2.363	7.459	14.289	5.421
all_vacuum	$(0, 0)$	50	0.147	0.515	0.447	0.947	6.798	2.558
all_vacuum	$(100, 0)$	50	0.154	0.364	0.416	0.999	4.075	1.956
all_vacuum	$(0, 20)$	50	0.174	0.411	0.396	1.002	6.913	2.299

Table 1: σ values from Geant4 + reconstruction ensemble; σ in mm, σ_θ in mrad. Robust half-width $(p_{84} - p_{16})/2$. Source: `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §4.

2.3 Gap to design specifications

The realistic-air spatial σ_y(PDC2) sits roughly an order of magnitude away from the 1 mm hardware design figure, and only the all-vacuum ablation reaches the design floor. Table 2 summarises the comparison; angular and momentum entries follow the same structure.

Quantity	Design / commissioning	Our current state	Source
Position σ at PDC	1 mm	$\sim$ 13 mm (all-air) / $\sim$ 7.5 mm (A=vac) / $\sim$ 1.0 mm (all-vac)	Kobayashi 2013 §3.4[1] + memo §4[6]
Angular σ	2 mrad	5.4–9.2 mrad (σ_{θ_y}, all-air & `beamline_vacuum`) / 2.0–2.6 mrad (all-vacuum)	Kobayashi 2013 §3.4[1] + memo §4[6]
RI commissioning $\Delta p/p$	1/1500	not measured in this ablation	Kobayashi 2013[1]
Proton-mode design $\Delta p/p$	$\sim$ 1/700	not measured in this ablation	RIBF-TAC05 (2005)[5]

Table 2: PDC reconstruction performance: design / commissioning targets vs. our current Geant4 + reconstruction state. Position and angular rows compare against the Kobayashi 2013 §3.4 hardware specification; the two $\Delta p/p$ rows are not yet propagated end-to-end in our ablation, so we have σ_y(PDC2) but no single $\Delta p/p$ figure to report.

2.4 Why this matters for the science floor

The PDC2 σ_y entries above are not the end of the story: they propagate backwards through the Runge–Kutta tracker and the SAMURAI magnetic field map to a target-momentum uncertainty $\sigma_{p,\text{target}}$, which is the reconstruction science floor for any experiment in which the target sits inside the magnet and the outgoing proton is measured downstream (target $\rightarrow$ magnet $\rightarrow$ PDC). This document does not re-derive $\sigma_{p,\text{target}}$ propagation values; the per-method numerical breakdown lives in the RK error-analysis deep-dive memo[9].

3 PDC reconstruction reference frame

3.1 Hardware foundations

The PDC1/PDC2 hardware specification is documented across four canonical sources: Kobayashi 2013 §3.4[1], the RIKEN internal `Detector-PDC.pdf` tech note[2], the SAMURAI 2012 construction proposal[4], and the older RIBF-TAC05 (2005) construction proposal[5]. The proton-arm magnet is the SAMURAI superconducting H-shape dipole described by Sato et al., providing a 7 Tm bending power, a maximum field of 3 T, an 80 cm pole gap and a 2 m pole face, with the field map computed in OPERA-3d / TOSCA[10]. The drift cells follow the Walenta geometry with an 8 mm drift distance, arranged as five planes (U/V/X/U/V) at $\pm 45°/0°$ over a 1700×800 mm^2 active area, filled with Ar + 25% iC_4H_{10} (or alternatively Ar + 50% C_2H_6) and read out via $\sim$810 channels[2]. The chambers sit at $\sim 59°$ on the proton arm with the design targets of 1 mm position resolution, 2 mrad angular resolution, and 0.8% relative momentum resolution, with commissioning RI runs reaching $\Delta p/p \approx 1/1500$[1] and a proton-mode design specification of $\Delta p/p \approx 1/700$[5]. The signal chain feeds an ASD shaper into AMT-VME TDCs at 0.78 ns/ch[5], where the AMT chip is the TDC ASIC originally described by Arai[11]. The boundary between the dipole-cavity vacuum (Region A) and the downstream air volume (Region B) is the SAMURAI Exit Window described by Shimizu et al.: a Kevlar foil of 280 µm backed by a Mylar foil of 75 µm[12]. The matching offline reconstruction pipeline is the RIKEN ROOT-based drift-chamber analysis described by Otsu et al.[13]. Together these four canonical sources cover the hardware geometry and design specifications, but none of them publishes a head-to-head proton-arm reconstruction performance paper; that gap is the motivation for §3.2 (algorithm spine) and §3.3 (performance achieved).

[9] See `docs/reports/reconstruction/momentum_reco/rk/rk_error_analysis_deep_dive_20260426_zh.pdf` for $\sigma_{p,\text{target}}$ propagation under MC, Profile, Laplace, MCMC, and MC-PDC.

[10] H. Sato et al., IEEE Trans. Appl. Supercond. **23**, 4500308 (2013), DOI: 10.1109/TASC.2012.2237225.

[11] Y. Arai, Nucl. Instr. Meth. A **453**, 365 (2000), DOI: 10.1016/S0168-9002(00)00658-6.

[12] Y. Shimizu et al., Nucl. Instr. Meth. B **317**, 739–742 (2013), DOI: 10.1016/j.nimb.2013.08.059.

[13] H. Otsu et al., Nucl. Instr. Meth. B **376**, 175 (2016), DOI: 10.1016/j.nimb.2016.02.056.

Item	Value	Source
PDC location	$\sim 59°$ on proton arm	Kobayashi 2013 §3.4
Active area	1700×800 mm^2	Detector-PDC tech note
Drift cell	Walenta, 8 mm drift	Detector-PDC tech note
Plane configuration	U/V/X/U/V at $\pm 45°/0°$	Detector-PDC tech note
Primary gas	Ar + 25% iC_4H_{10}	Detector-PDC tech note
Alternative gas	Ar + 50% C_2H_6	Detector-PDC tech note
Channels	~ 810	Detector-PDC tech note
Readout TDC	AMT-VME, 0.78 ns/ch	RIBF-TAC05 2005 + Arai 2000
Magnet type	superconducting H-dipole	Sato 2013
Bending power	7 Tm	Sato 2013
Max field	3 T	Sato 2013
Gap	80 cm	Sato 2013
Pole face	2 m	Sato 2013
Field map source	OPERA-3d / TOSCA	Sato 2013
ExitWindow	Kevlar 280 μm + Mylar 75 μm	Shimizu 2013
Design position σ	1 mm	Kobayashi 2013 §3.4
Design angle σ	2 mrad	Kobayashi 2013 §3.4
Design $\Delta p/p$	0.8%	Kobayashi 2013 §3.4
Commissioning RI $\Delta p/p$	1/1500	Kobayashi 2013
Proton mode $\Delta p/p$ design	$\sim 1/700$	RIBF-TAC05 2005

Table 3: PDC1/PDC2 + magnet + ExitWindow hardware specifications. Sources cited inline; full bibliographic entries in §7.

3.2 Algorithm spine

The published reference reconstruction chain for the SAMURAI proton arm follows a four-stage spine. **(i) Hits to track segments.** Drift times measured by the AMT-VME TDCs are converted to drift distances and combined into a straight-line track in each PDC plane; this drift-time-to-distance and per-plane line-fit pipeline is the RIKEN ROOT-based drift-chamber analysis described by Otsu et al.[13], deployed within the SAMURAIANA / `anaroot` framework (cite by repo path `anaroot/`). **(ii) Back-tracking through the magnet.** The PDC-plane track is propagated backwards through the SAMURAI superconducting H-dipole using a Runge–Kutta integrator over the OPERA-3d / TOSCA field map published by Sato et al.[10]; the field map itself is peer-reviewed, but the back-tracking algorithmic recipe is documented only in PhD theses (see point iii) and not in any peer-reviewed paper. **(iii) Optional Kalman smoothing.** A Kalman-filter smoother on the PDC tracks is the standard refinement, but *PDC-side Kalman smoothing has no peer-reviewed publication; the full methodology lives only in PhD theses.* The closest published paper, Santamaria et al., Nucl. Instr. Meth. A **905**, 138–148 (2018)[14], applies a Kalman filter to the MINOS TPC, not the PDCs. The PDC-side methodology is recorded across four PhD theses: Santamaria 2015 (Univ. Paris-Saclay)[15], Kahlbow 2019 (TU Darmstadt)[3], Lehr ca. 2023 (TU Darmstadt)[16], and Storck 2023 (TU Darmstadt)[17]. **(iv) Output.** The propagated state at the target plane yields the target-frame momentum vector $\vec{p}_{\text{target}}$ used downstream. As context, several MINOS+SAMURAI groups generate their training and validation data using the NPTool simulation framework of Matta et al., J. Phys. G **43**, 045113 (2016)[18], but NPTool is a simulation framework, not the reconstruction path itself.

[14] C. Santamaria et al., Nucl. Instr. Meth. A **905**, 138–148 (2018), DOI: 10.1016/j.nima.2018.07.053.

[15] C. Santamaria, PhD thesis, Univ. Paris-Saclay (2015), HAL tel-01231191, `https://hal.science/tel-01231191`.

[16] C. Lehr, PhD thesis, TU Darmstadt (2023).

[17] S. Storck, PhD thesis, TU Darmstadt (2023), TUprints 23768, `https://tuprints.ulb.tu-darmstadt.de/23768/`.

[18] A. Matta et al., J. Phys. G **43**, 045113 (2016), DOI: 10.1088/0954-3899/43/4/045113.

Our internal pipeline lives in `libs/analysis_pdc_reco/` (see also `smsimulator5.5/CLAUDE.md` for the repo-level guide) and follows the same Runge–Kutta back-tracking spine, with multiple selectable RK fit modes; one concrete example is `--rk-fit-mode fixed-target-pdc-only` as exercised in the MS ablation memo[6] §3. An alternative neural-network backend is provided under `scripts/reconstruction/nn_target_momentum/`, and an RK geometry filter (`libs/qmd_geo_filter/`, with the matching skill `skills/smsim-rk-geometry-filter/`) prunes obviously off-acceptance trajectories before the fit. We do not re-derive the per-method numerical performance here: the full status, including MC, Profile, Laplace, MCMC, and MC-PDC variants, is reported in the RK status memo at `docs/reports/reconstruction/momentum_reco/rk/rk_reconstruction_status_20260416_` and the deep-dive[9]. A detailed algorithm-by-algorithm comparison between the literature spine and our internal pipeline is deferred to §4.2 (Methodology gap).